ASTROPHYSICS LABORATORY
Observational Astrophysics 2025

I) Basic notions in Observational astrophysics

a) Properties of the astronomical objects

- Magnitude; Absolute magnitude
- Magnitude and distance: ly, parsec
- Temperature and colour
- Size

b) Preparing astronomical observations

- Visibility of the target: coordinates and time
- Telescope
- Detector and data acquistion

II) Observations: data acquisition

III) Data reduction and analysis

Detector system

Instruments at the CIGeoE
telescope + filters+CCD camera



Detector system

Instruments at the CIGeoE
telescope + filters+CCD camera



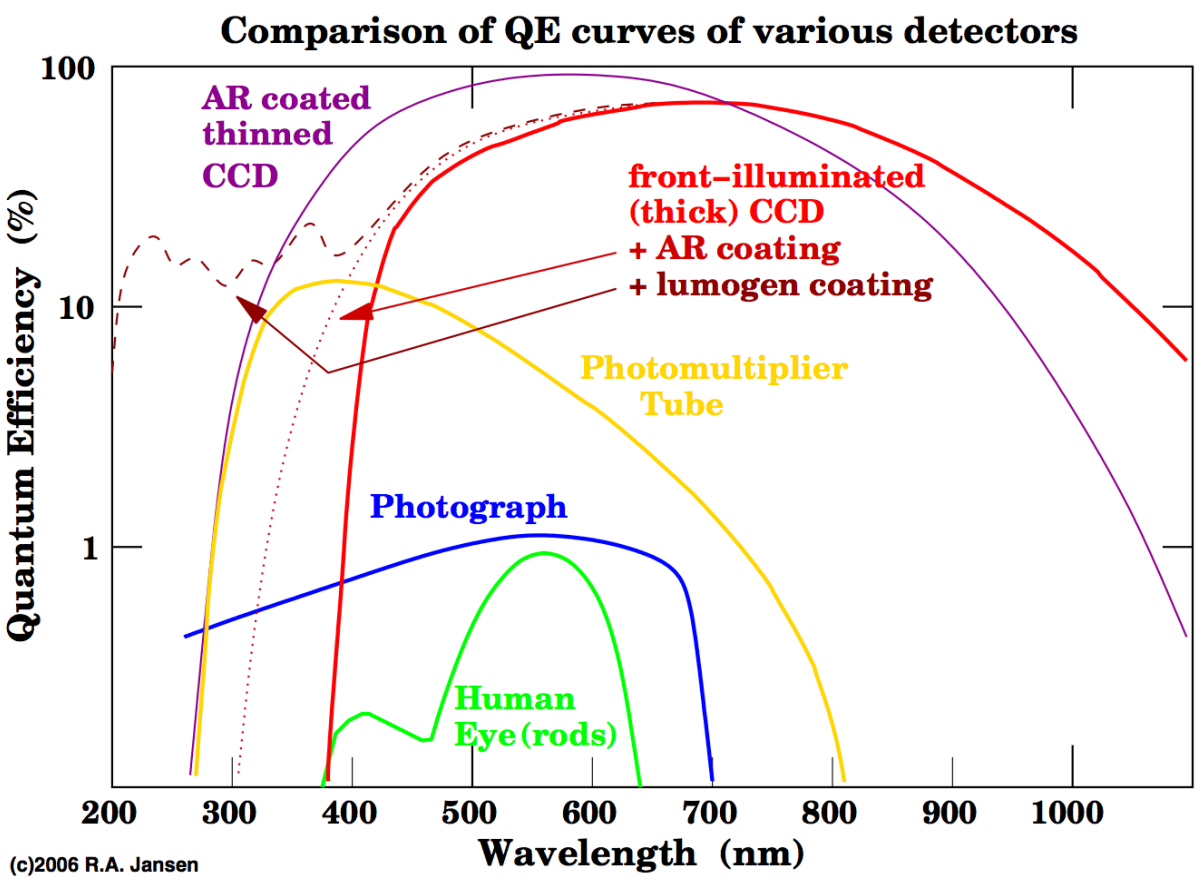


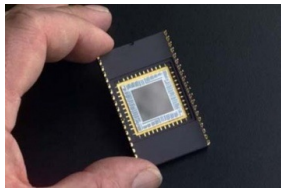
CCD Camera

MORAVIAN G2-1600

https://www.gxccd.com/art?id=441&cat=52&lang=409

CCD







https://www.gxccd.com/download/G2%20Mark%20II%20Manual.pdf

Characteristics of the CCD detector

Charge-coupled devices (CCDs) are silicon-based integrated circuits consisting of a dense matrix of photodiodes that operate by converting light energy in the form of photons into an electronic charge. Electrons generated by the interaction of photons with silicon atoms are stored in a potential well and can subsequently be transferred across the chip through registers and output to an amplifier. The schematic diagram illustrated in Figure 1 shows various components that comprise the anatomy of a typical CCD.

When a ultraviolet, visible, or infrared photon strikes a silicon atom resting in or near a CCD photodiode, it will usually produce a free electron and a "hole" created by the temporary absence of the electron in the silicon crystalline lattice. The free electron is then collected in a potential well (located deep within the silicon in an area known as the depletion layer), while the hole is forced away from the well and eventually is displaced into the silicon substrate. Individual photodiodes are isolated electrically from their neighbors by a channel stop, which is formed by diffusing boron ions through a mask into the p-type silicon substrate.

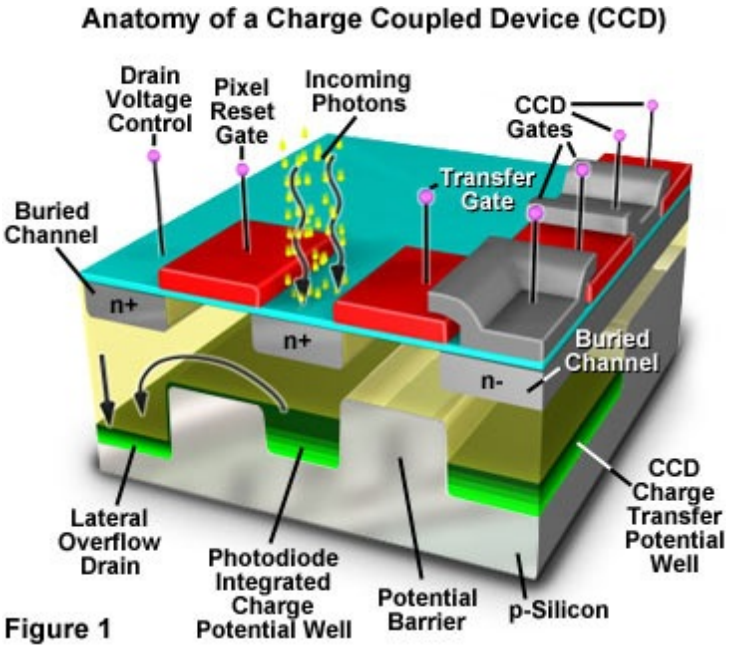
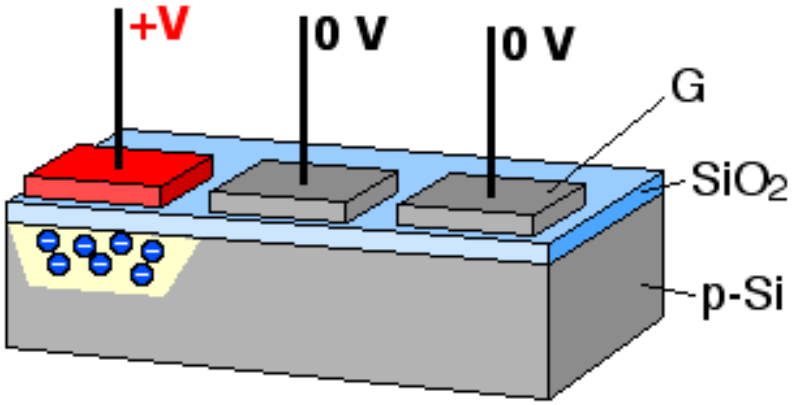


Figure 1

https://hamamatsu.magnet.fsu.edu/articles/ccdanatomy.html
https://hamamatsu.magnet.fsu.edu/tutorials/

Characteristics of the CCD detector

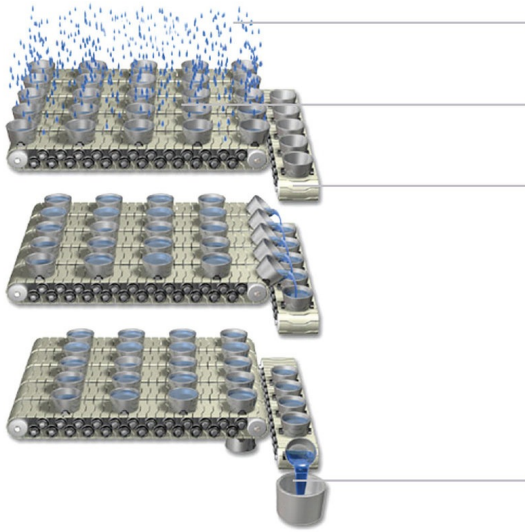


The charge packets (electrons, blue) are collected in potential wells (yellow) created by applying positive voltage at the gate electrodes (G). Applying positive voltage to the gate electrode in the correct sequence transfers the charge packets.

https://en.wikipedia.org/wiki/Charge-coupled_device

CCD readout

Conveyor belt analogy



Photons
CCD

Serial
register

Output
amplifier

Bias: thermal electrons at zero exposure time

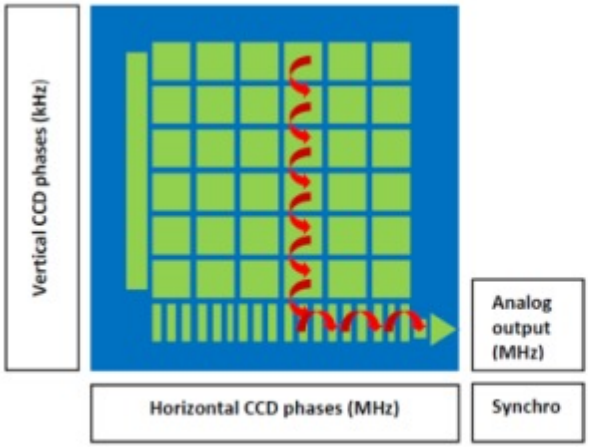
Dark current: thermal electrons are indistinguishable from photo-electrons

- > CCDs are cooled!

readout time: takes time do download the signal, and there is

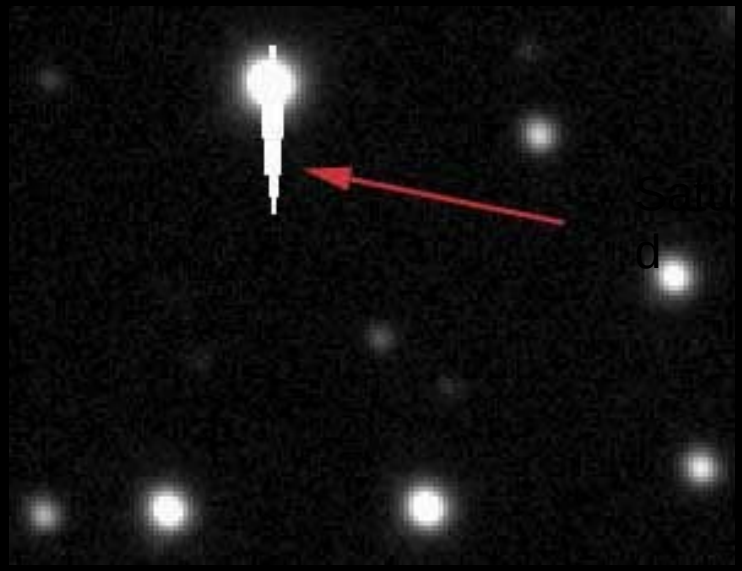
readout noise

Characteristics of the CCD detector



CCD = Photon-to-electron conversion (analog)

Characteristics of the CCD detector



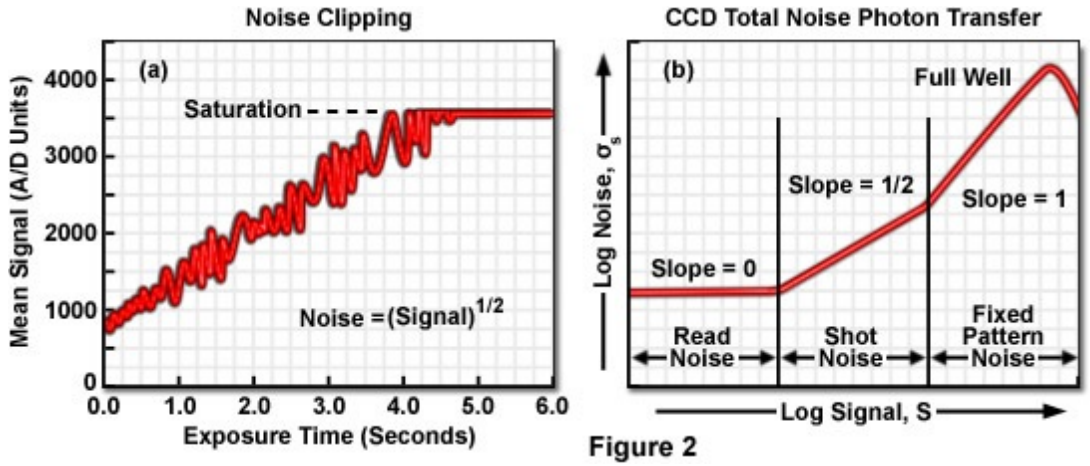

Noise Clipping
(a)
Saturation
Noise = (Signal)^1/2
Mean Signal (A/D Units)
4000
3000
2000
1000
0
0.0
1.0
2.0
3.0
4.0
5.0
6.0
Exposure Time (Seconds)
CCD Total Noise Photon Transfer
(b)
Full Well
Slope = 1/2
Slope = 1
Slope = 0
Log Noise, σs
Read Noise
Shot Noise
Fixed Pattern Noise
Log Signal, S

Figure 2

Quantum efficiency (QE)

effectiveness to produce electrons from photons

ADU: Analog to digital units

Each pixel has some counts of ADU.

Gain

Amplification of initial number of pixel photoelectrons into final ADU counts:
gain = # of electrons/ #counts [e^-/ADU]

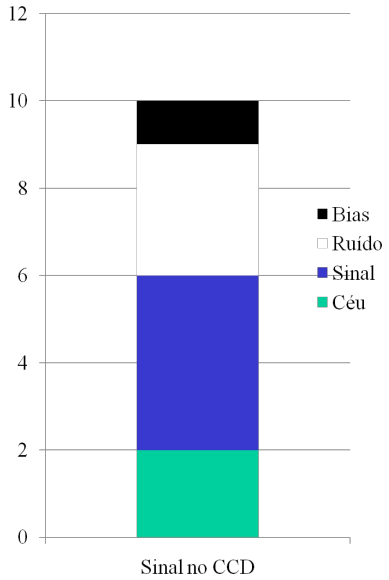
This is chosen so that maximum ADU matches the Full well depth (maximum charge a pixel can hold).

Credi for the slide:
S. Gonzalez-Gaitán, LA@IST 2023

How to extract a signal from a CCD image?

- **Signal of the astronomical object**
- **sky+atmosphere**
- **CCD noise**

  Bias(t=0s) Dark (t, T)



Other problems

- ***Bad* pixels**
- **Inhomogeneities in CCD sensitivity**
- **Cosmic rays**

- ***Bad pixels mask***
- **Correction *Flat***
- ***CR correction***

- ***Ignore bad pixels***
- **Flat illuminated image**
- ***Various exposures***

Characteristics of the CCD detector

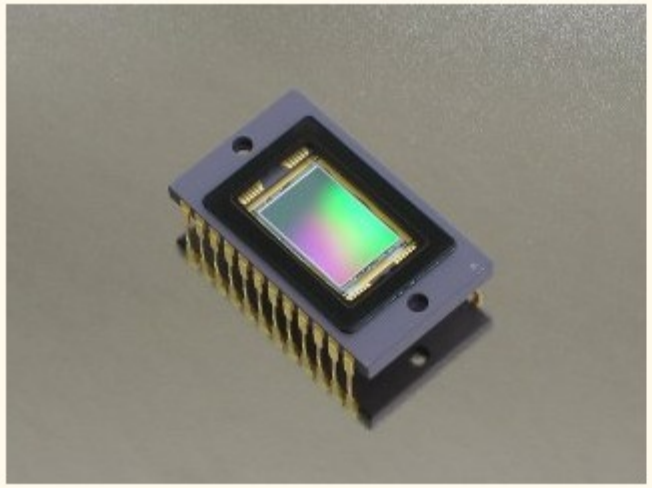
Model G2-1600

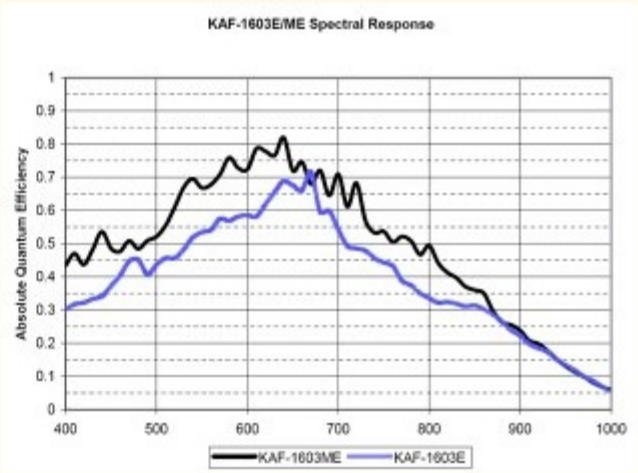
G2-1600 model uses 1.6 MPx OnSemi KAF-1603ME NABG Full-Frame CCD chip.

Moravian

Resolution	1536 (H) × 1024 (V) pixels
Pixel size	9 µm (H) × 9 µm (V)
Imaging area	13.8 mm (H) × 9.2 mm (V)
Full well capacity	~100,000 e^-
Output node capacity	~220,000 e^-
Anti-blooming gate	No
Dark current	1 e^-/s/pixel at 0 °C
Dark signal doubling temperature	6.3 °C

KAF-1603ME CCD chip specifications





KAF-1603ME CCD chip and its Quantum Efficiency

https://www.gxccd.com/art?id=441&cat=52&lang=409

Characteristics of the CCD detector

Gain	1.5 e^-/ADU (1 × 1 binning)
	2.0 e^-/ADU (other binnings)
System read noise	14 e^- RMS (Low Noise mode)
	16 e^- RMS (Preview mode)
Full frame download	2.6 s (Low Noise mode)
	1.8 s (Preview mode)

G2-1600 electronics specifications

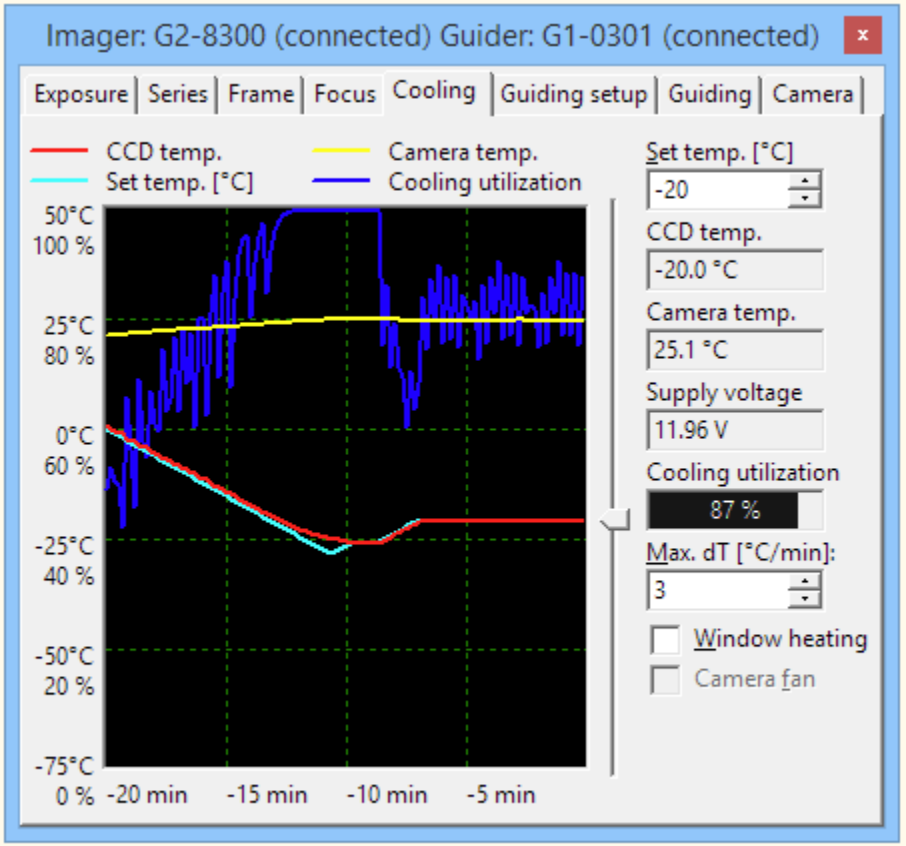
https://www.gxccd.com/art?id=441&cat=52&lang=409

Characteristics of the CCD detector

Chip Cooling

CCD chip cooling	Thermoelectric (Peltier modules)
TEC modules	Two stages
Max. Δ T	50 °C below ambient maximum
	45 °C below ambient typical
Regulation precision	0.1 °C
Hot side cooling	Forced air cooling (fan)

Chip cooling specifications



Cooling tab of the CCD Camera control tool in the SIPS software package

It is not recommended to cool the chip to the maximum temperature difference, else the camera cannot guarantee temperature stability when the ambient air temperature rises. It is usually practical to set the temperature so the cooling utilization varies around 90%. This provides enough reserve in cooling power to keep the CCD temperature even if the ambient temperature rises several degrees Celsius.

https://www.gxccd.com/art?id=441&cat=52&lang=409

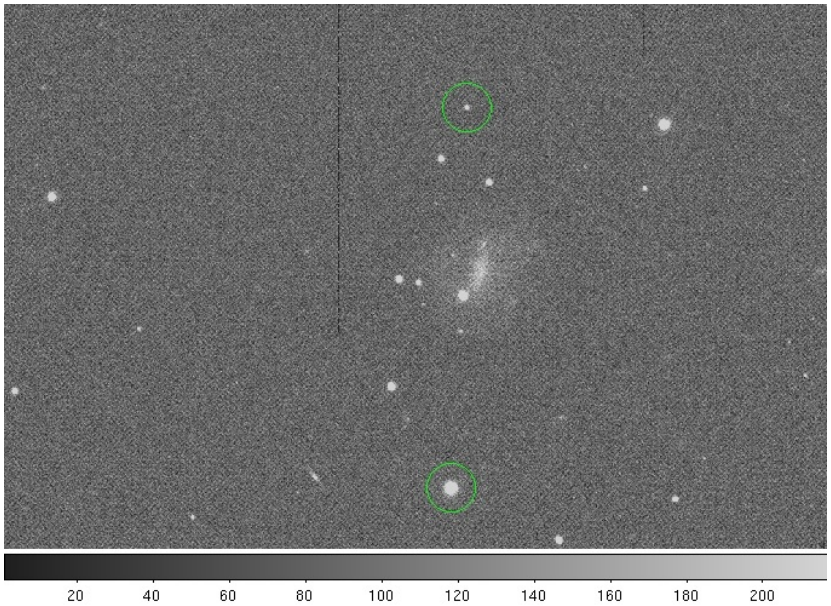
Characteristics of the CCD detector

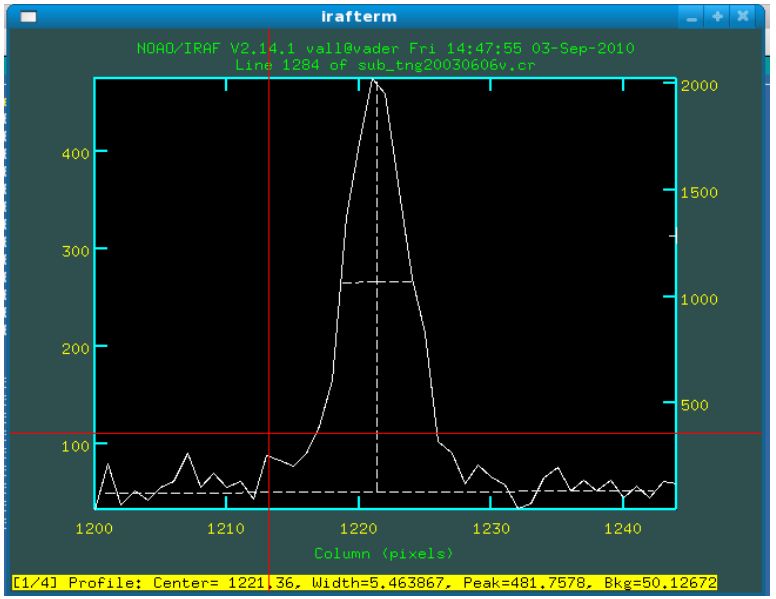
Mechanical specifications

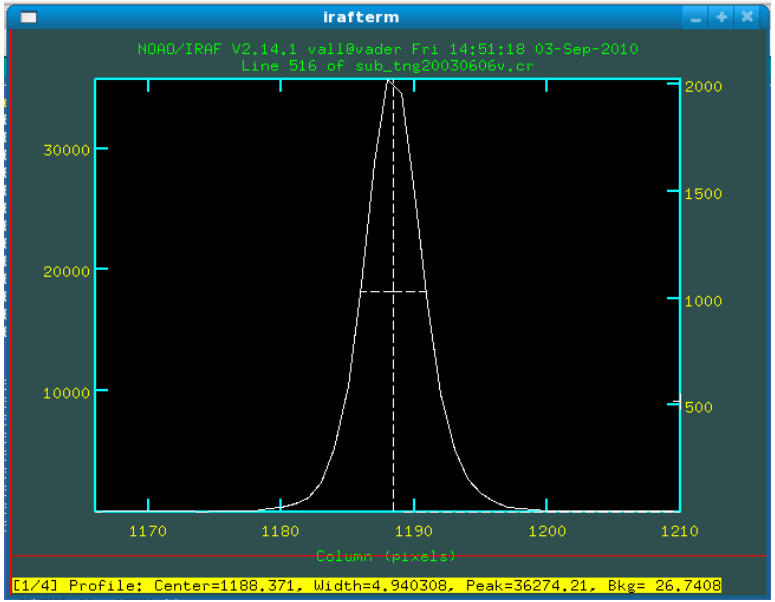
Internal mechanical shutter	Yes, blade shutter
Shortest exposure time	0.1 s
Longest exposure time	Limited by chip saturation only
Internal filter wheel	5 positions for 1.25" threaded filter cells or for D31 mm unmounted filters
	6 positions for 1" or D26.5 mm unmounted filters
Head dimensions	114 mm × 114 mm × 77.5 mm (with internal filter wheel)
	114 mm × 114 mm × 65 mm (without filter wheel)
Back focal distance	29 mm (with internal filter wheel)
	16.5 mm (without filter wheel)
	33.5 mm (with external filter wheel)
Camera head weight	1.15 kg (with internal filter wheel)
	1.05 kg (without filter wheel)
	1.70 kg (with “XS” external filter wheel)
	1.95 kg (with “S” external filter wheel)

https://www.gxccd.com/art?id=441&cat=52&lang=409

Data analysis and reduction

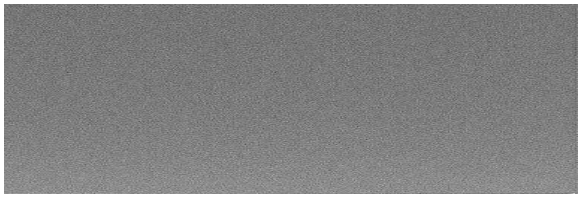




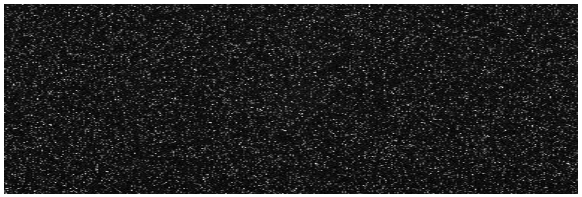


Small and big stars have the same fwhm

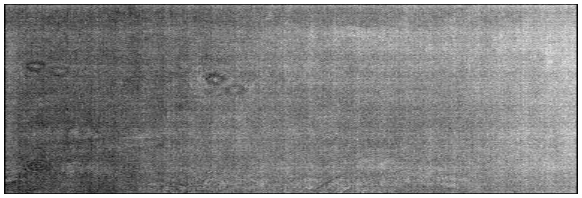
Detector, Data analysis and Data reduction



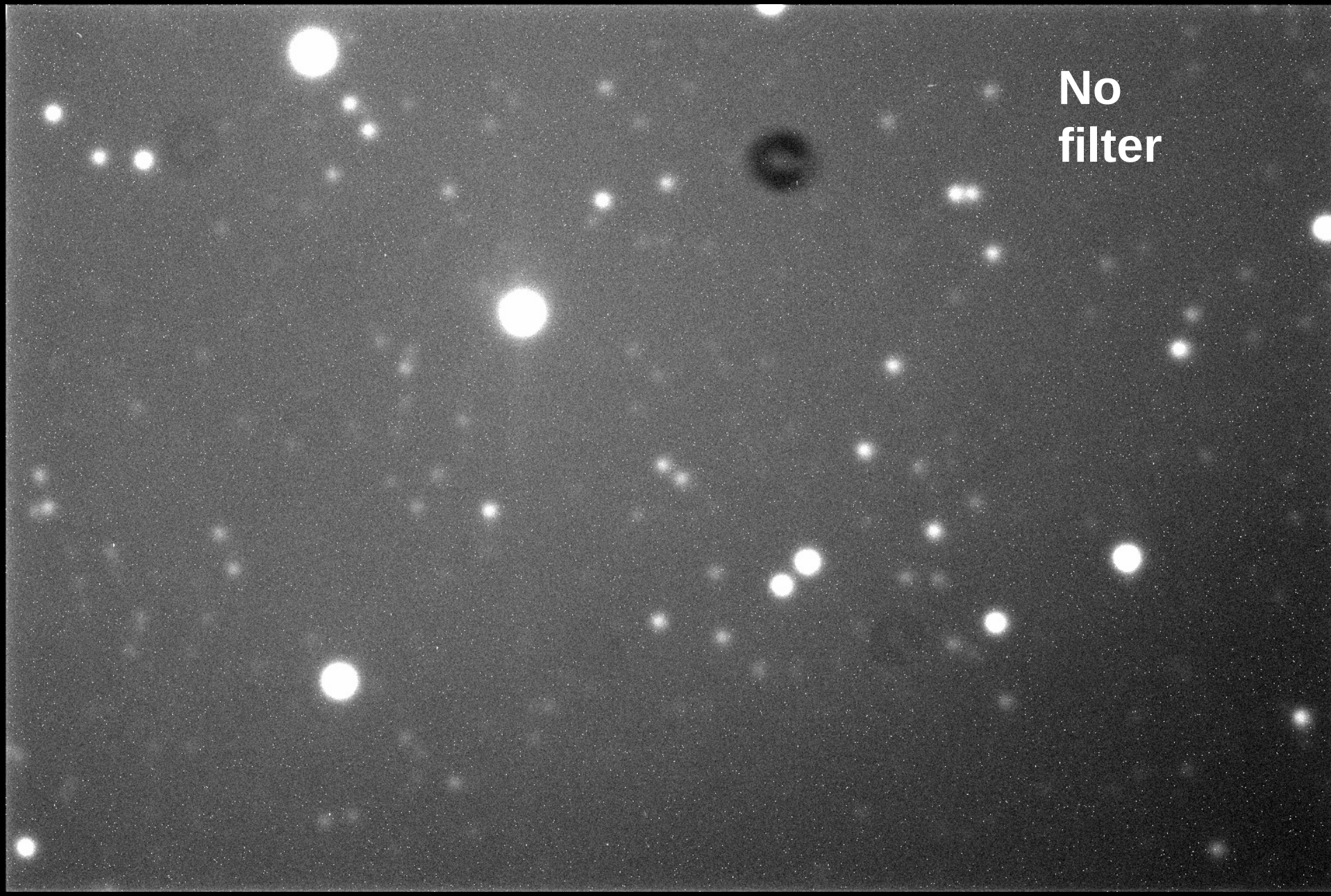
Bias (-)



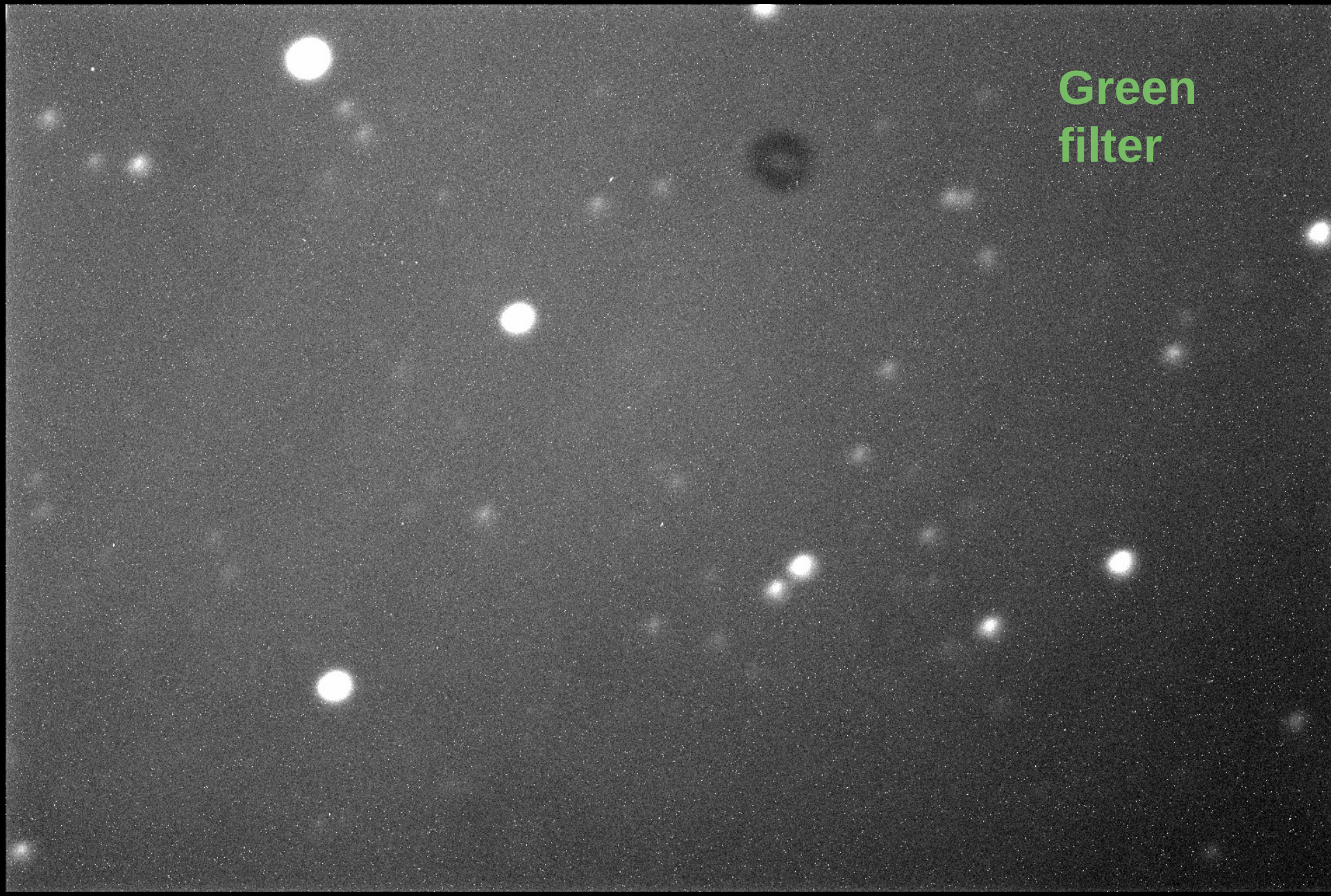
Dark (-)



Flat (/)

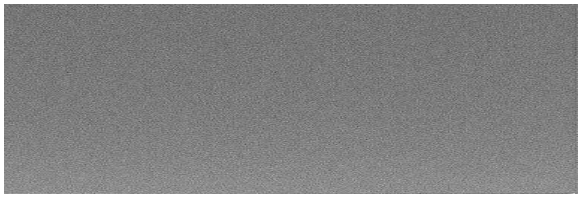

No
filter


Blue
filter

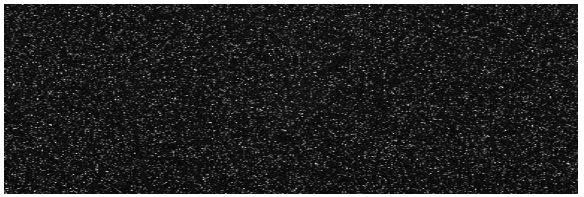

Green
filter


Red
filter

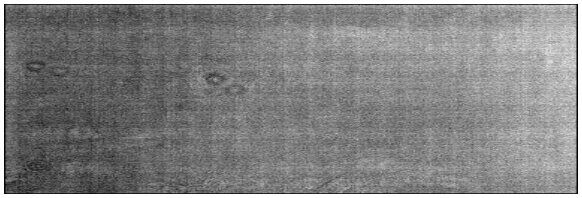
Detector, Data analysis and Data reduction



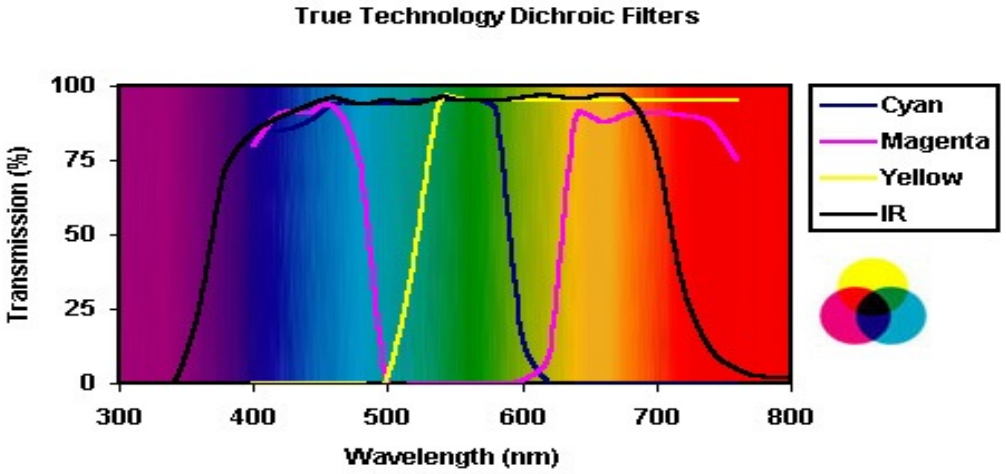
Bias (-)



Dark (-)



Flat (/)



CCD sources of noise

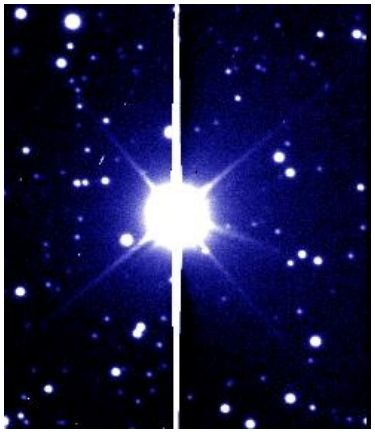
Linearity of response

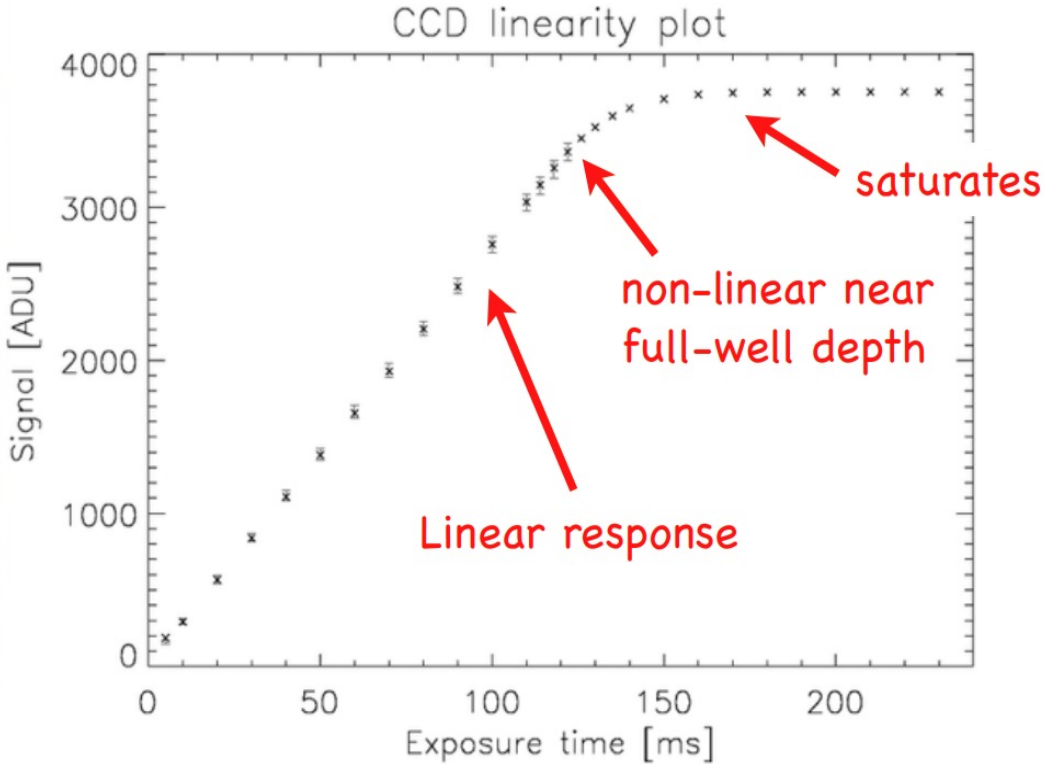
amount of photons is linearly proportional to signal over large range

Saturation

max. number of electrons in potential well of pixel is reached

non-linear response





SIGNAL $S = N_* \times Q_E \times t$

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CCD sources of noise

- **Shot noise or Poisson noise**:
- statistical fluctuations of discrete particles of charge/light.
- At high N_*, target counts [e-/s/pix]:

$$\sigma_* \approx \sqrt{N_* \times Q_E \times t}$$



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CCD sources of noise

- **Shot noise or Poisson noise**: statistical fluctuations of discrete particles of charge/light. At high N_*, target counts [e-/s/pix]:

$$\sigma_* \approx \sqrt{N_* \times Q_E \times t}$$

- **Sky or background noise**: Poisson noise from the sky, with the count rate S [e-/s/pix] (depends for example on moon phase or light pollution): $\sqrt{S \times Q_E \times t}$

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Moon brightness

Dark sky:
22mag/arcsec^2

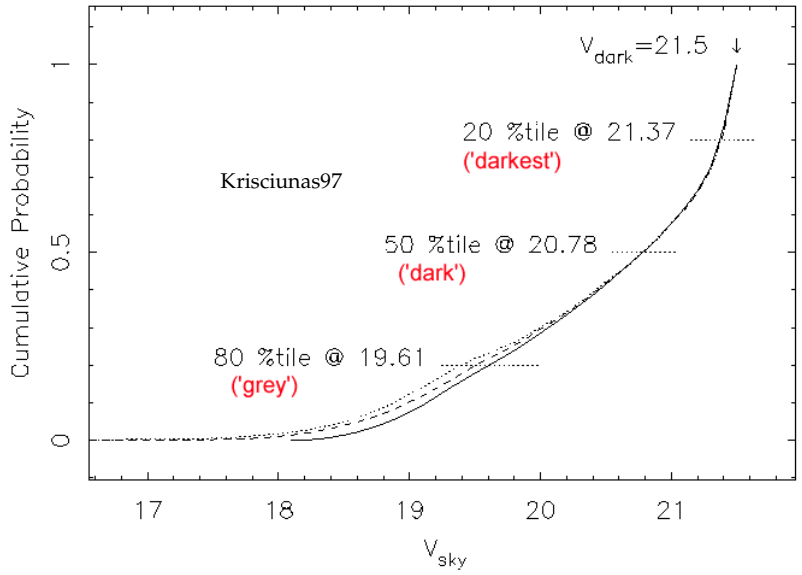
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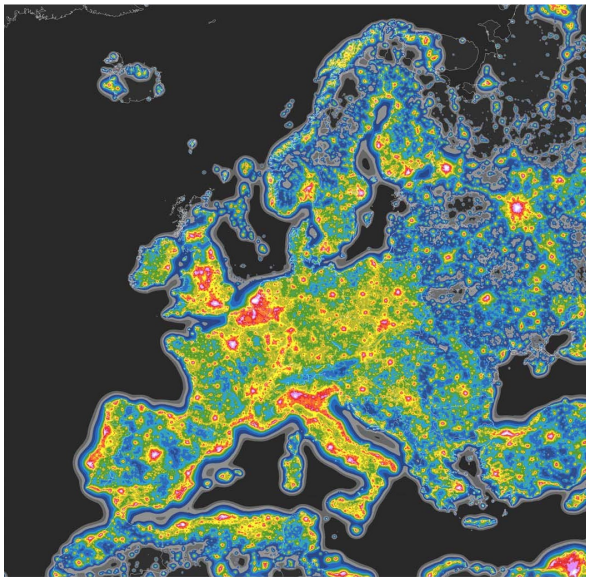
$$\sigma_* \approx \sqrt{N_* \times Q_E \times t}$$

- **Sky or background noise**: Poisson noise from the sky, with the count rate S [e-/s/pix] (depends for example on moon phase or light pollution):

$$\sqrt{S \times Q_E \times t}$$



Moon brightness



Lisbon ~10 x dark sky

Dark sky: 22mag/arcsec^2

CCD sources of noise

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- **Sky or background noise**: Poisson noise from the sky, with the count rate S [e-/s/pix] (depends for example on moon phase or light pollution): $\sqrt{S \times Q_E \times t}$
- **Dark current:** thermal fluctuations: some electrons jump potential well of CCD

  - depends on temperature
  - is proportional to time.
  - With I_D [e/s/pix] as the dark current:

$$\sqrt{I_D \times t}$$

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CCD sources of noise

- **Shot noise or Poisson noise**: statistical fluctuations of discrete particles of charge/light. At high N_*, target counts [e-/s/pix]:

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$$\sqrt{S \times Q_E \times t}$$

- **Dark current:** thermal fluctuations: some electrons jump potential well of CCD

$$\sqrt{I_D \times t}$$

- **Readout noise (RON)**: electronic noise [e/pix] from imperfect electronic devices. It depends on temperature, read-out speed, amplifier…

$$N_r$$

$$\text{total noise} = \sqrt{\sigma_*^2 + \sigma_{sky}^2 + \sigma_{dark}^2 + RON^2}$$

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CCD sources of noise

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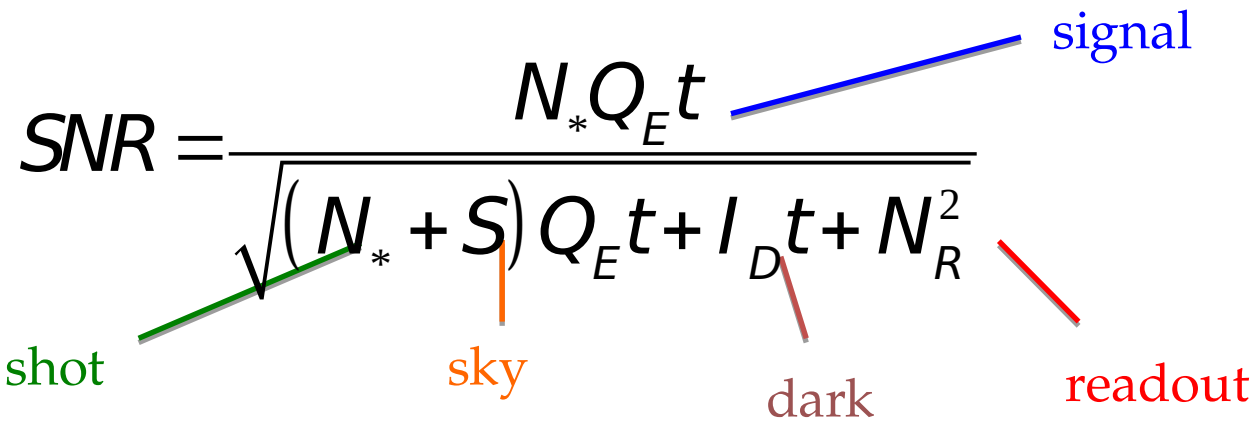
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Credi for the slide:
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CCD

signal to noise ratio in one pixel



Note that N_*,S and QE are a function of wavelength!!

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Hands-on!

CIGeoE TELESCOPE



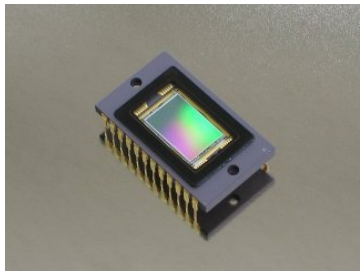
Telescope CELESTRON characteristics:

- ❑ Schmidt-Cassegrain telescope
- ❑ Aperture diameter: 35.4cm (14inch)
- ❑ Focal length: 4000mmm
- ❑ Equatorial mount

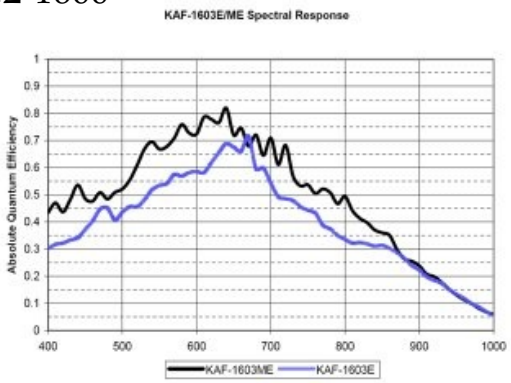
Camera MORAVIAN characteristics:

- ❑ Size: 1536 x 1024 pixel
- ❑ Pixel size: 9 x 9 µm
- ❑ Readout time: 2.1s
- ❑ Readout noise: 16e- rms
- ❑ Dark current: 1e-/s/pix
- ❑ Gain: 1.5 e-/ADU





G2-1600



Hands-on!

CIGeoE TELESCOPE



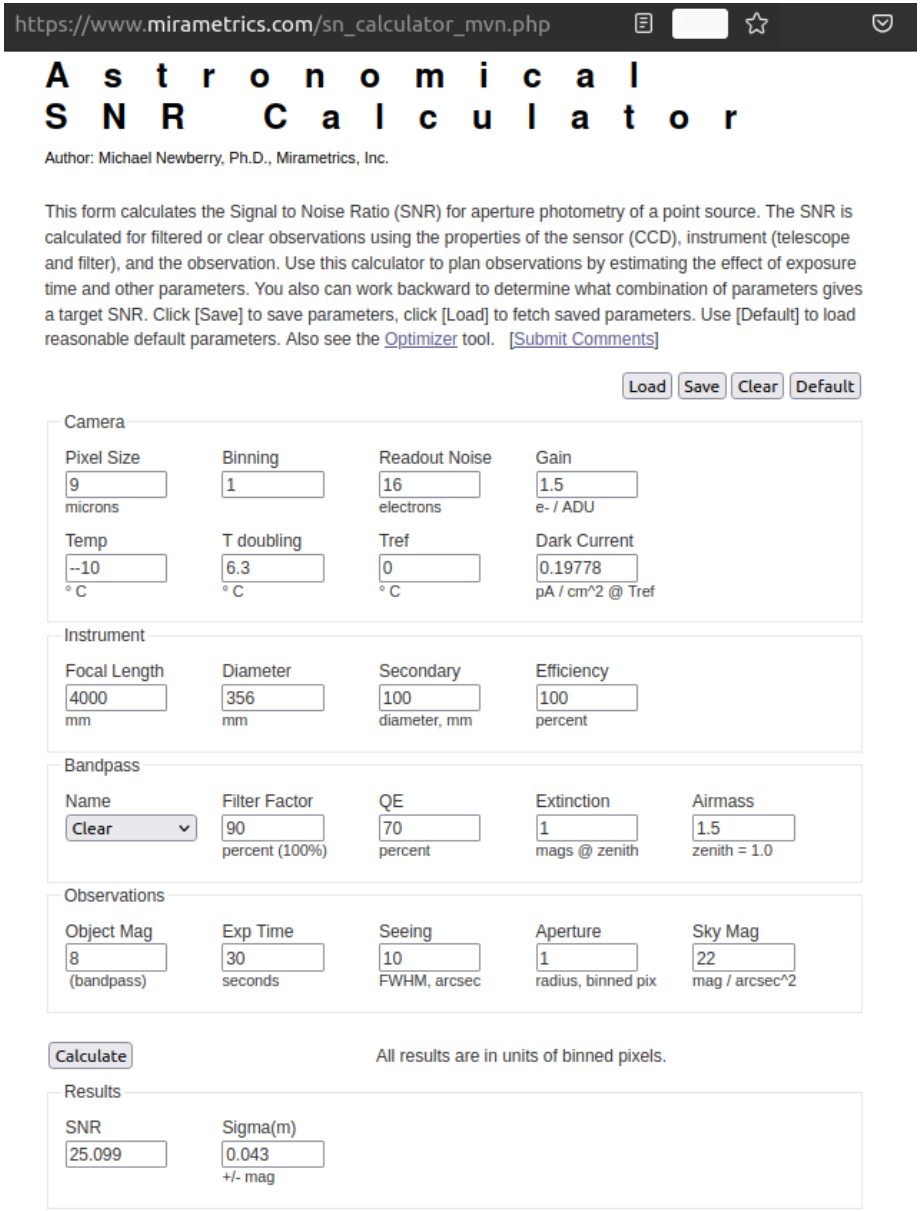
1) **How long do we need to expose at CIGeoE with the G2-1600 camera to obtain S/N = 10;50;100 for:**

- Deneb(V=1.3mag), SU Cyg (V=7mag)
- Dumbbell – M27 (V=7.5, A=8’.0x5’.6)

Use of exposure time calculator:
https://www.mirametrics.com/sn_calculator_mvn.php

Camera info:
https://www.gxccd.com/art?id=321

use the following specifications:



Newberry, M. V. 1991, PASP 103, 122-130

Hands-on!

CIGeoE TELESCOPE



2) **How would the images look like in the field of view of the G2-1600 camera to obtain for same objects:**

- Deneb(V=1.3mag), SU Cyg (V=7mag), VW Hya (V=12mag)
- Dumbbell – M27 (V=7.5, A=,8’.0x5’.6)

Use of field-of-view calculator
https://www.skyatnightmagazine.com/astronomy-field-view-calculator/

Camera info:
https://www.gxccd.com/art?id=321

Hands-on!

CIGeoE TELESCOPE



Answer:

https://www.x2r.uk/fovcalc.php

2R

FOV Calculator / Telescope simulator

Targets	Mode	Image	Options	Help

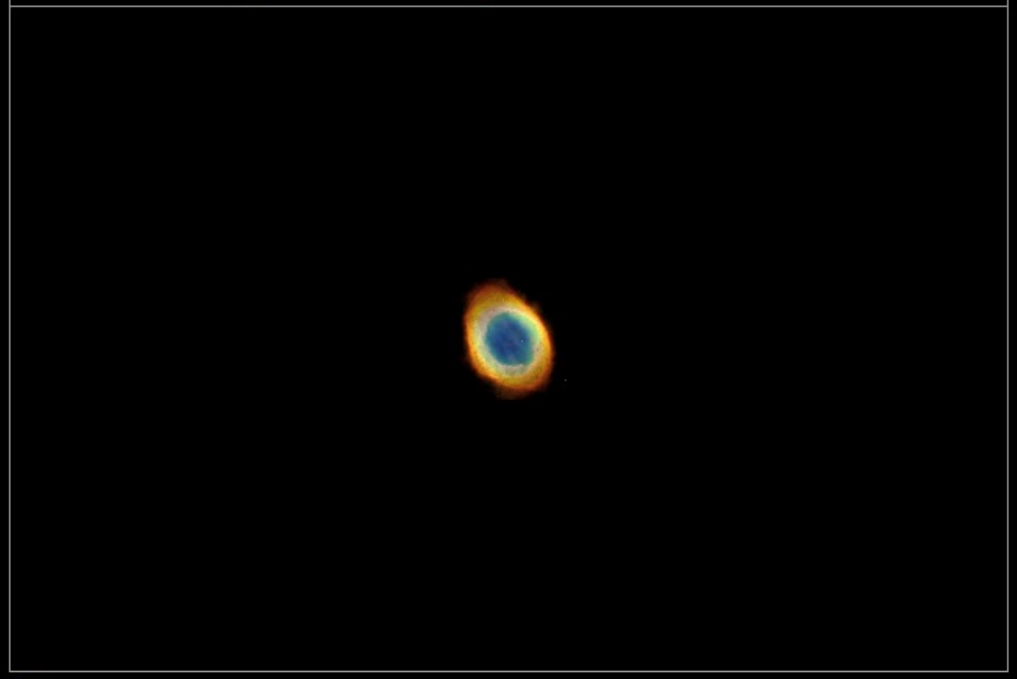
Celestron
C14-A XLT
- Aperture: 355.6mm
- Focal length: 3910mm
- Focal ratio: f/11.0
- Barlow/Focal reducer: 1x

Celestron
Camera model...
- Pixel size: 9 µm
- Image size: 1536 x 1024
- Sensor size: 13.8 x 9.2mm
- Binning: 1x1

Results

FOV: 12.15' x 8.10'
Resolution: 0.47"/pixel
Area: 98.5 sq'

Focal length: 3910mm
Focal ratio: f/11.0



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Image credit: NASA / JPL / STScI

Database: 564 telescopes, 542 cameras, 482 eyepieces, 83 binoculars. Add more? Last updated 21 January 2021

CIGeoE TELESCOPE

WRITE an observing proposal

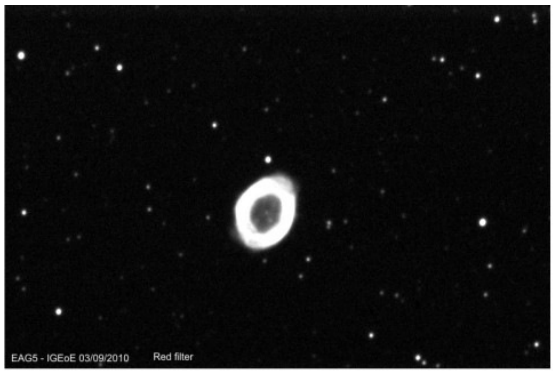
OBSERVE and acquire data

REDUCE the data and

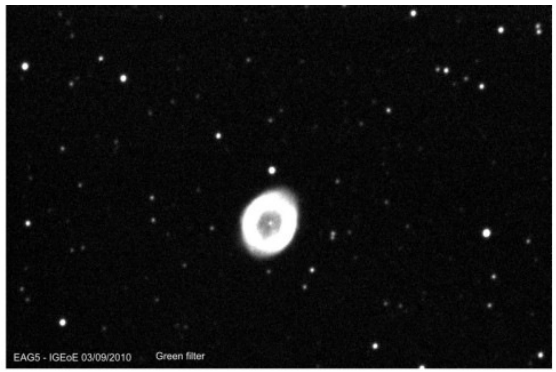
ANALYSE the results

HANDS-ON

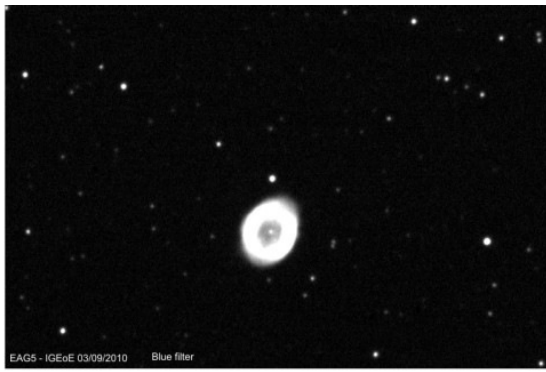
Data analysis and reduction



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